Tutorial Sheet 4

IIT Delhi – Abu Dhabi

Course Code: AENL226

Course Name: Power Electronics and Power Systems

Complex Power and Phasor Analysis

Unless otherwise stated, all voltage and current phasors are RMS quantities. Use the passive sign convention for complex power.

1. Question

Two sinusoidal voltage sources are connected through a series impedance. The source voltages are

$$E_1 = 120\angle 0^\circ \text{ V}, \qquad E_2 = 100\angle -20^\circ \text{ V},$$

and the impedance connecting the two sources is

$$Z = 2 + j6 \ \Omega.$$

The current I is defined positive from Source 1 toward Source 2.

(a) Determine the current I in rectangular and polar form.

(b) Determine the active and reactive power absorbed by each source.

(c) Determine the complex power absorbed by the connecting impedance.

(d) Verify conservation of complex power.

(e) Draw the phasor diagram showing E_1, E_2, $V_Z = E_1 - E_2$, and I.

2. Question

Two AC voltage sources are connected through a series resistive-capacitive impedance. The source voltages are

$$E_1 = 110\angle 10^\circ \text{ V}, \qquad E_2 = 125\angle -15^\circ \text{ V},$$

and

$$Z = 1 - j4 \ \Omega.$$

The reference current I flows from Source 1 toward Source 2.

(a) Determine the line current I.

(b) Determine the complex power absorbed by each source.

(c) Determine the active and reactive power absorbed by the connecting impedance.

(d) Identify which source is supplying active power and which source is absorbing active power.

(e) Verify conservation of complex power.

(f) Draw the phasor diagram showing E_1, E_2, $V_Z = E_1 - E_2$, and I.